

Ziheng Cheng

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EDUCATION

Ph.D. in Industrial Engineering and Operations Research

Sep, 2024-May, 2028 (Expected)

UC Berkeley, CA, USA

B.S. in Mathematics

Sep, 2020-Jun, 2024

Peking University, Beijing, China

RESEARCH INTEREST

Generative AI (LLM & Diffusion Model), Reinforcement Learning, Machine Learning Theory, Game Theory, Statistics

SELECTED PUBLICATIONS & MANUSCRIPTS

(* stands for equal contribution)

- [1] How RL Reshapes LLM Reasoning: Transferability, Coverage, and Scaling Laws
Ziheng Cheng*, Yixiao Huang*, Hanlin Zhu, Somayeh Sojoudi
(preprint)
- [2] Periodic Weak Spots: Phase Sensitivity from Chunked KV-Cache Compression
Xingyu Zhu*, Yi (Luke) Pu*, **Ziheng Cheng***, Ang Lv*, Jing Liu, Lexing Ying, Yiyuan Ma, Xin Dong
(preprint)
- [3] Actor-Critic Learning for Extended Mean Field Control with Deterministic Policies
Ziheng Cheng, Xin Guo, Huyên Pham, Yufei Zhang
(preprint, under review of Operations Research)
- [4] Multi-Mask Diffusion Language Models for Few-Step Generation
Sijin Chen*, YINUO Ren*, Heyang Zhao, **Ziheng Cheng**, Quanquan Gu, Lexing Ying
(COLM 2026)
- [5] Multi-Objective Learning for Diffusion Models: A Statistical Theory under Semi-Supervised Learning
Ziheng Cheng*, Yixiao Huang*, Hanlin Zhu, Haoran Geng, Somayeh Sojoudi, Jitendra Malik, Pieter Abbeel, Xin Guo
(ICML 2026)
- [6] Deterministic Policy Gradient for Reinforcement Learning with Continuous Time and State
Ziheng Cheng, Xin Guo, Yufei Zhang
(preprint, under review of Mathematics of Operations Research)
- [7] OVERT: A Benchmark for Over-Refusal Evaluation on Text-to-Image Models
Ziheng Cheng*, Yixiao Huang*, Hui Xu, Somayeh Sojoudi, Xuandong Zhao, Dawn Song, Song Mei
(NeurIPS 2025)
- [8] Provable Sample-Efficient Transfer Learning Conditional Diffusion Models via Representation Learning
Ziheng Cheng, Tianyu Xie, Shiyue Zhang, Cheng Zhang
(NeurIPS 2025)
- [9] Convergence of Distributed Adaptive Optimization with Local Updates
Ziheng Cheng, Margalit Glasgow
(ICLR 2025)
- [10] Kernel Semi-Implicit Variational Inference
Ziheng Cheng*, Longlin Yu*, Tianyu Xie, Shiyue Zhang, Cheng Zhang
(ICML 2024)
- [11] Momentum Benefits Non-IID Federated Learning Simply and Provably
Ziheng Cheng*, Xinmeng Huang*, Pengfei Wu, Kun Yuan
(ICLR 2024)
- [12] Joint Graph Learning and Model Fitting in Laplacian Regularized Stratified Models
Ziheng Cheng*, Junzi Zhang*, Akshay Agrawal, Stephen Boyd
(preprint, under review)

RESEARCH EXPERIENCE

LLM Pretraining & Sparse Attention Mechanism

Jan, 2026 -

Advisor: Lexing Ying, Stanford University & Byedance Seed

- Studied sparse attention architecture design in LLM pretraining - Established efficient KV-cache compression algorithms	
Reinforcement Learning & Stochastic Control	May, 2025 -
Advisor: Prof. Xin Guo, UC Berkeley	
- Studied reinforcement learning in continuous time with deterministic policy. - Established a mathematical rigorous framework for SDE-driven reinforcement learning for the first time. - Achieved the state-of-the-art performance in various benchmarks.	
Safety Alignment in Generative AI	Oct, 2024 - May, 2025
Advisor: Prof. Song Mei, UC Berkeley	
- Studied the safety landscape in generative AI, especially on text-to-image models. - Constructed an automatic pipeline based on LLM to generate large scale evaluation benchmarks. - Analyzed over-refusal phenomenon and safety-utility tradeoff in T2I models for the first time.	
Distributed Optimization & Federated Learning Theory	Mar, 2023 - Oct, 2023
Advisor: Prof. Tengyu Ma, Stanford University; Prof. Kun Yuan, Peking University	
- Studied the benefits of local iterations to reduce communication in distributed setting. - Proposed a distributed adaptive optimization algorithm based on gradient-clipping and Adam. - Achieved the convergence result of Adam in distributed setting for the first time. - Proved that momentum can accelerate the convergence of classic FL algorithms without additional assumption.	
Diffusion Model & Learning Theory & Sampling Algorithm	May, 2022 - Oct, 2024
Advisor: Prof. Cheng Zhang, Peking University	
- Studied sample complexity theories of transfer learning diffusion models. - Studied particle-based variational inference methods to accelerate MCMC sampling.	
INDUSTRY EXPERIENCE	
Bytedance Seed, San Jose, CA, USA	Jan, 2026 -
Research Intern	
Manager: Lexing Ying	
Project: Large Language Model Pre-training	
Microsoft Research Asia, Beijing, China	Oct, 2023 - May, 2024
Research Intern	
Manager: Zhong Li, Jiang Bian	
Project: Data Selection for Efficient Training LLM	
AWARDS AND HONORS	
Awards	
- Honorable Mention in Alibaba Global Mathematics Competition - Bronze Medal in S.-T. Yau College Student Mathematics Contest - Meritorious Winner in Mathematical Contest in Modeling	2022, 2023, 2024 2022 2022
Honors	
- IEOR Faculty Fellowship (top-1 in UC Berkeley IEOR) - May-Fourth Scholarship (top scholarship in Peking University, 0.5%) - National Scholarship (top 0.2% nation-wide) - Merit Student of Peking University	2026 2023 2021 2021-2023